

AFL Data Foundations — Data Dictionary & Target Definitions

Qasim | BSSE 2022 (2022-SE-49) | Week 3, Day 1 Assignment | Due: 14 Sept 2026

Dataset Inventory

Match results (2012-2018, akareen/AFL-Data-Analysis, sourced from AFL Tables) and player-match performance (2012-2018, 4CDA/AFL-ANN, sourced from stoney71's Kaggle AFL stats export) — both downloaded live from GitHub, no API keys used. 7 seasons, 18 teams, 1,240 players.

Table	Rows	Cols	Grain	Primary Key	Foreign Keys
matches	1,447	16	One row per match	match_id	—
player_stats	63,712	39	One row per player per match (team perspective)	Player + match_id	match_id, team
teams	18	2	One row per team	team_id	—

Data-quality findings: 1 missing fixture (2017 EF, Port Adelaide v West Coast — 44 unjoinable player rows); 88 player rows with null match metadata (Collingwood's 2018 QF/EF squads — a source-site scraping gap); the raw match source's "attendance" column actually holds kickoff time-of-day, not a crowd figure, and is 72% null (renamed, excluded from analysis); 1 fully-blank match row dropped (2015 file); no duplicate rows in either raw source; team names already consistent across both sources — no harmonization needed.

Target Definitions

Target	Definition	Formula	Level
home_win	1 if home team wins, 0 otherwise	(home_score > away_score).astype(int)	Match
top_disposal_getter	Player with most disposals in a match	player_stats.groupby('match_id')['Disposals'].idxmax()	Player-game
top_goal_kicker	Player with most goals in a match	player_stats.groupby('match_id')['Goals'].idxmax()	Player-game
fantasy_points	Composite AFL-Fantasy-style performance score	3×Kicks + 2×Handballs + 3×Marks + 4×Tackles + 1×Hitouts + 1×FreesFor – 3×FreesAgainst + 6×Goals + 1×Behinds	Player-game

Classification, not regression: home_win is modeled as win/loss rather than margin — margin is noisier game-to-game (blowouts and close finishes compress to one number a model must match exactly), while win/loss is the outcome that matters for tipping/ladder purposes and what the form/rest/ladder features below are hypothesized to predict directly.

Feature Dictionary (features_v1.parquet)

Feature	Description	Window	Source Columns
last_5_avg_score	Avg. points scored over the last 5 games	5 matches (shifted)	home/away_score
days_rest	Days since the team's previous match	1 match (diff)	match_date
win_streak_entering	Consecutive wins (+) or losses (–) entering the match	Running (shifted)	win
last_5_avg_player_output	Avg. team-wide player fantasy_points over last 5 games	5 matches (shifted)	player_stats.fantasy_points
ladder_position	Team rank by competition points accumulated before this round	Season-to-date (shifted)	win
h2h_win_rate	Win rate vs. this specific opponent in all prior meetings	All-time (prior only)	team, opponent, win

All rolling/cumulative features are computed with **shift(1) before rolling/cumsum** — sorted by team and date, so a feature for any given match uses only games strictly before it. Saved as features_v1.parquet, 1,447 rows x 28 columns.

Split Logic & Prediction Ceiling

time_based_split(df, holdout_season=2018) trains on all seasons strictly before the holdout season and tests only on it — every model this week must call this exact function so results are comparable. Train: 1,240 matches (2012-2017). Test: 207 matches (2018). A random row-level split would leak future information: a team's rolling form/ladder/H2H features are built from its own past games, so scattering games from the same season across train/test would let a model implicitly see a team's rest-of-season form or a rematch result when training on an earlier meeting — information a real predictor would never have. It would also leak season-specific effects (a new coach, a rule change) from test-season games mixed into training.

Realistic ceiling: AFL outcomes are inherently noisy (injuries, umpiring variance, a bouncing oval ball), so a model that predicted every game correctly would signal leakage, not skill. Home-ground advantage alone gives a ~57-58% baseline (see notebook Figure 2); published AFL tipping models combining form, ladder position, and head-to-head typically reach **~65-75% match-winner accuracy** on held-out seasons — the realistic target for this week's models.